### 2.9.5 Bubbled AND Gate

The Logic Circuit of Bubbled AND Gate

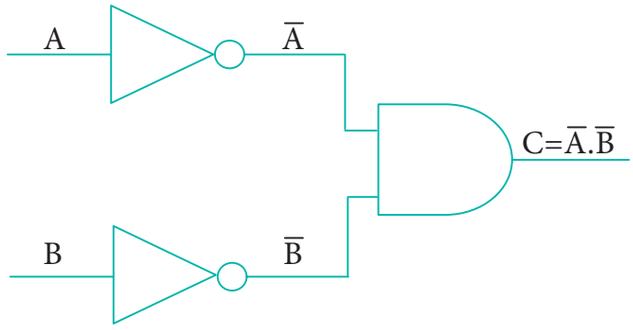



*Fig. 2.9 Logic circuit of Bubbled AND Gate*

In the above circuit, invertors on the input lines of the AND gate gives the output as

$$C = (\overline{A} \cdot \overline{B})$$

This circuit can be redrawn as the bubbles on the inputs, where the bubbles represent inversion.

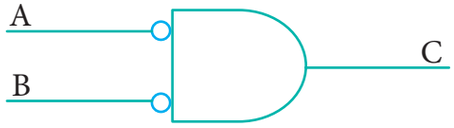



*Fig. 2.10 Logic Symbol of Bubbled AND Gate*

We refer this as bubbled AND gate. Let us analyse this logic circuit for all input possiblities.

If A = 0 and B = 0 C=($\overline{0}$.$\overline{0}$) = 1.1 = 1

If A = 0 and B = 1 C=($\overline{0}$.$\overline{1}$) = 1.0 = 0

If A = 1 and B = 0 C=($\overline{1}$.$\overline{0}$) = 0.1 = 0

If A = 1 and B = 1 C=($\overline{1}$.$\overline{1}$) = 0.0 = 0

Here the truth table is

| Input | | Output |
|---|---|---|
| A | B | C |
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

You can see that, a bubbled AND gate produces the same output as a NOR gate. So, You can replace each NOR gate by a bubbled AND gate. In other words the circuits are interchangeable.

Therefore

$$\overline{(A + B)} = \overline{A} \cdot \overline{B}$$

Which establishes the De Morgan's first theorem.

### 2.9.6 NAND Gate

The NAND gate operates an AND gate followed by a NOT gate. It acts in the manner of the logical operation "AND" followed by inversion. The output is "false" if both inputs are "true", otherwise, the output is "true". In otherwords the output of the NAND gate is 0 if and only if both the inputs are 1, otherwise the output is 1. The logic circuit of NAND gate is

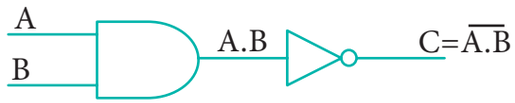



*Fig. 2.11 Logic Circuit of NAND Gate*

The logical symbol of NAND gate is




*Fig. 2.12 Logic Symbol of NAND Gate*

The output of the NAND gate is

$$C = \overline{(A \cdot B)}$$

Read this as "C" equals NOT of A AND B" or "C" equals the complement of A AND B".

For example if both the inputs are 1



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